



Course Information

Course Name: Advanced Computer Systems

Course ID: CIS 570

Semester: Fall 2026

Room: Dion 110

Instructor(s) Information

Name: Dr. Adnan El-Nasan (*Prof. NAS*)

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Office Hours (in Dion 302-F): **Tue, Thur** (3:00 to 4:00 PM); **Wed** (from 2:00 to 4:00)

Teaching Assistant: Wenhao Wang

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Office Hours: Mon 3:00 to 4:00 PM (in Dion 309), Fri 11:00 to 12:00 AM (in Dion 309)

Course Description

In-depth treatment of current computer systems, with performance issues at the center of an analytical approach. The course explores operating system software and the interrelation between architecture and system software. Advanced topics of compiling, assembly, linking and loading of high-level language software are included. The course treats mechanisms of IO and the memory hierarchy, various features of traditional machines, advanced features of modern machines such as RISC and multi-processor machines, and file systems and networked and distributed systems such as inter- and intra-nets. Throughout, performance issues are at the center of an analytical approach.

Prerequisite:

CIS 314; CIS 273 and CIS 370; or equivalent knowledge

Course Credits: 3 credits

Required Text: All three books below are excellent resources for the course but for your reference, I will provide you with notes that are based on the material in these books. As a CIS major, I recommend that you have a good and contemporary book on Computer Architecture (the first book below will be my recommendation as I might assign chapter readings from it).

1.	<u>"Computer Architecture: A Quantitative Approach", by John L. Hennessy and David A. Patterson, Morgan Kaufmann Publishers, 2025 (7th ed.)</u> ISBN: Paperback: 9780443154065 eBook: 9780443154072
2.	"Computer Organization and Design: The Hard-ware/Software Interface", by David Patterson and John Hennessy, Morgan Kaufmann, 2013 (5 th ed.)
3.	Computer System Design and Architecture", by Vincent Heuring, Harr Jordan, Pearson Prentice Hall, 2004, (2 nd ed.)

Course Objectives

By the end of the course, students will:

- Understand the operation of modern microprocessors at an architectural level.
- Understand the operation of memory and I/O subsystems and their relation to overall system performance.
- Understand the benefits of multiprocessor/many-processor systems and the difficulties in designing and utilizing them.
- Gain familiarity with simulation techniques used in computer architecture.

Course Schedule

Tentative Week	Topics to be Covered
1	Instruction Set Architecture (ISA) Basics
2	MIPS
3	ISA Implementation
4	Basic Pipelining:
5	Dependencies and Hazards
6	Loop Unrolling
7	Branch Prediction
8, 9	Advance Pipelining: Dynamic Scheduling,
10	Multithreading
11	Memory Hierarchy
12	Virtual Memory and
14	Cache Optimization
15	Review

Syllabus Changes:

This syllabus is a guide, and every attempt is made to provide an accurate overview of the course. However, circumstances and events may make it necessary for the instructor to modify the syllabus during the semester and may depend, in part, on the progress, needs, and experiences of the students. Changes to the syllabus will be made with notice.

Evaluation Plan and Grading System:

Evaluation will be based on the following components:

- 30% Assignments (quizzes, reading assignments, HWs, class discussions and participation, ...).
- 20% Midterm Exam.
- 20% Final Project
- 30% Final exam (**Comprehensive** and around **Dec. 11**).

Grading:

LETTER CODE	PERCENTAGE
A+	95 - 100
A	90 - 95
A-	87 - 89
B+	84 - 86
B	80 - 83
B-	77 - 79

LETTER CODE	PERCENTAGE
C+	74 - 76
C	70 - 73
C-	67 - 69
D+	64 - 66
D	60 - 63
F	< 60

For any grades not mentioned in the above tables, we'll follow the UMass Dartmouth grading system as defined here: [https://catalog.umassd.edu/content.php?catoid=62&navoid=5015#Grades and Grading System](https://catalog.umassd.edu/content.php?catoid=62&navoid=5015#Grades_and_Grading_System)

Late Assignments:

Each assignment will have its due date listed so please make sure to complete all assignments on time. When an extension is needed, please contact the TA as early as possible and state your reasons. Extensions will be granted only for hard conditions and when we are given enough time in advance.

Late submission of assignments is accepted within **5 days from the due date** with a penalty of **20% for every late day**. No submission will be allowed:

- 5 days after the due date (without prior permission).
- If the assignment is relevant to an upcoming exam or quiz.

Class Policies:

AI Use Policy:

AI tools are prevalent, and their application is revolutionizing engineering practices. Consequently, it is crucial to attain a level of discernment that enables us to validate their outputs and make informed decisions.

This course allows you to use these tools to help in the learning process and in developing the fundamental knowledge needed for the course based on the following rules:

- AI is **encouraged** for brainstorming, literature review, debugging, calculation checks, practice problem generation and visualization/formatting.
- AI is **allowed**, with proper documentation, for analyzing complex problems, drafting reports, creating initial designs, and analyzing data. The documentation should include a list the AI tool(s) used, your prompts, how you validated the output, what changes you made and why, and what you eventually learned.
- AI, in any form, is **prohibited** on exams and quizzes, oral presentations and defenses, and any individual assessments explicitly marked "No AI."

AI and its tools can be used to facilitate the learning of the course material but is **prohibited, in any hardware or software form**, during assessments (e.g., **exams or quizzes, lab submissions...etc.**). **If in doubt, please ask the course instructor.**

Attendance:

Students are expected to attend their classes and ask questions. In the event that an illness requires a student to miss more than the equivalent of two weeks of classes (i.e., four classes for a class meeting twice a week for 75 minutes), it is important for the student to communicate the circumstances surrounding their absences to the instructor immediately to be given instructions on how to make up missed material.

Academic Calendar: <https://www.umassd.edu/online/support/academic-calendar/>

Exams and Quizzes:

- Exams and Quizzes will be held **in person** and will cover material from the lectures and textbook.
- If a student has a legitimate excuse for missing an exam, she/he must get approval from the instructor **in advance**.
- Some quizzes are **not announced in advance** and will cover the **material** discussed in **previous lectures**.
- **No makeup Quizzes** at all (some quizzes will not be announced in advance).
- All exams and quizzes are **closed book and closed notes**.

UMass Dartmouth Student Academic Integrity Policy:

All UMass Dartmouth students are expected to maintain high standards of academic integrity and scholarly practice. The University does not tolerate academic dishonesty of any variety, whether as a result of a failure to understand required academic and scholarly procedure, or as an act of intentional dishonesty. In case of a violation a student will be subject to proper actions in accordance to the University Academic Integrity Policy:

<http://www.umassd.edu/policies/activepolicylist/academicaffairs/academicintegritypolicyandreportingform/>

On-campus Internet Access:

Spaces will be available on campus with access to high-speed internet, including the library. Designated seats in rooms on campus may be assigned to students who require access to high-speed internet.

Center for Access and Success

In accordance with university policy, if you have a documented disability and require accommodations to obtain equal access to this course, please meet with the instructor at the beginning of the semester and provide the appropriate paperwork from the [Center for Access and Success](#). The necessary paperwork is obtained when you bring proper documentation to the Center.

University Academic Policies

These policies are also available in the student handbook on the University website - umassd.edu

- [Information on Incompletes](#)
- [Student Academic Integrity](#)
- [Definition of Credit Hour](#)
- [Course Withdrawal](#)
- [Grade Appeal](#)
- [Attendance Policy](#)
- [Academic Calendar](#)
- [Title IX and Sexual Assault/Harassment](#)

Technical Support

24/7 email, live chat, and phone support for myCourses is available online. For help with myCourses and other UMass Dartmouth technologies, [Please contact CITS](#).